

T0-Theory: Complete Derivation of All Parameters Without Circularity

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Abstract

This documentation presents the derivation of the parameters of T0-theory in logical order, with the aim of excluding circularity. The presentation shows how the fine structure constant $\alpha = 1/137$ is meant to follow from geometric principles without presupposing it. The derivation steps are documented explicitly so that the charge of circularity can be checked step by step. The early coupling-constant formulas of this document are classified as preliminary stages (see the classification at the end of the document).

Note on versions. The German and English versions of this document differ in structure and extent; earlier revisions shortened, extended or changed parts of both. Where they differ, the more extensive German version applies; sections contained only in this English version count as supplements.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

.1 Introduction

T0-theory takes the approach of treating fundamental physical constants not as arbitrary but as derivable from the geometric structure of three-dimensional space. The central claim is that the fine structure constant $\alpha = 1/137.036$ is not an empirical input but follows from spatial geometry.

To make the question of circularity checkable, we present here the derivation of the parameters in logical sequence, starting from geometric principles and without using experimental values except fundamental natural constants; where a measured value does enter at some point, this is noted in the text.

.2 The Geometric Parameter ξ

Derivation from Fundamental Geometry

The universal geometric parameter ξ consists of two fundamental components:

$$\xi = \frac{4}{3} \times 10^{-4} \quad (1)$$

The Harmonic-Geometric Component: 4/3 as the Universal Fourth

4:3 = the fourth – a universal harmonic ratio

The factor 4/3 is interpreted here as the **perfect fourth**, one of the basic harmonic intervals:

$$\frac{4}{3} = \text{Frequency ratio of the perfect fourth} \quad (2)$$

Just as musical intervals are universal:

- **Octave:** 2:1 (always, whether string, air column, or membrane)
- **Fifth:** 3:2 (always)
- **Fourth:** 4:3 (always)

These ratios are **geometric/mathematical**, not material-dependent.

Why is the fourth universal?

For a vibrating sphere:

- When divided into 4 equal "vibration zones"
- Compared to 3 zones
- The ratio 4:3 emerges

This is **pure geometry**, independent of material.

The harmonic ratios in the tetrahedron:

The tetrahedron contains BOTH fundamental harmonic intervals:

- **6 edges : 4 faces = 3:2** (the fifth)
- **4 vertices : 3 edges per vertex = 4:3** (the fourth)

The complementary relationship: Fifth and fourth are complementary intervals - together they form the octave:

$$\frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = 2 \quad (\text{Octave}) \quad (3)$$

The document reads this as a harmonic structure of space:

- The tetrahedron contains both fundamental intervals
- The fourth (4:3) and fifth (3:2) are reciprocally complementary
- The harmonic structure is self-consistent

Further appearances of the fourth in physics:

- Crystal lattices (4-fold symmetry)
- Spherical harmonics
- The sphere volume formula: $V = \frac{4\pi}{3}r^3$

Interpretation:

- **Parallel to Pythagoras:** "Everything is number and harmony"
- **Space itself** has a harmonic structure
- **Particles** are "tones" in this cosmic harmony

T0 theory thus interprets space as harmonically structured, with 4/3 (the fourth) as its basic signature. This interpretation motivates the choice of the factor 4/3; it is not a derivation in the strict sense.

The 10^{-4} Factor:

Step-by-Step QFT Derivation:

1. Loop Suppression:

$$\frac{1}{16\pi^3} = 2.01 \times 10^{-3} \quad (4)$$

2. T0-Calculated Higgs Parameters:

$$(\lambda_h^{(T0)})^2 \frac{(v^{(T0)})^2}{(m_h^{(T0)})^2} = (0.129)^2 \times \frac{(246.2)^2}{(125.1)^2} = 0.0167 \times 3.88 = 0.0647 \quad (5)$$

3. Missing Factor to 10^{-4} :

$$\frac{10^{-4}}{2.01 \times 10^{-3}} = 0.0498 \approx 0.05 \quad (6)$$

4. Complete Calculation:

$$2.01 \times 10^{-3} \times 0.0647 = 1.30 \times 10^{-4} \quad (7)$$

What yields 10^{-4} : It is the T0-calculated Higgs parameter factor $0.0647 \approx 6.5 \times 10^{-2}$ that reduces the loop suppression by factor 20:

$$2.01 \times 10^{-3} \times 6.5 \times 10^{-2} = 1.3 \times 10^{-4} \quad (8)$$

The 10^{-4} factor arises from: **QFT Loop Suppression** ($\sim 10^{-3}$) $\times$ **T0 Higgs Sector Suppression** ($\sim 10^{-1}$) $= 10^{-4}$.

.3 The Mass Scaling Exponent κ

From the fractal dimension follows directly:

$$\kappa = \frac{D_f^{\text{eff}}}{2} = \frac{2.973}{2} \approx 1.487 \quad (9)$$

This exponent determines the nonlinear mass scaling in T0-theory.

.4 Lepton Masses from Quantum Numbers

The masses of leptons follow from the fundamental mass formula:

$$m_x = \frac{\hbar c}{\xi^2} \times f(n, l, j) \quad (10)$$

where $f(n, l, j)$ is the geometric structure factor of the quantum numbers (not a frequency and not the base winding number $f = 1/\xi$):

$$f(n, l, j) = \sqrt{n(n+l)} \times \left[j + \frac{1}{2} \right]^{1/2} \quad (11)$$

For the three leptons we obtain:

- Electron ($n = 1, l = 0, j = 1/2$): $m_e = 0.511 \text{ MeV}$
- Muon ($n = 2, l = 0, j = 1/2$): $m_\mu = 105.66 \text{ MeV}$
- Tau ($n = 3, l = 0, j = 1/2$): $m_\tau = 1776.86 \text{ MeV}$

As claimed in this document, these masses are not empirical inputs but follow from ξ and quantum numbers (for the classification, see the end of the document).

.5 The Characteristic Energy E_0

The characteristic energy E_0 follows from the geometric mean of the electron and muon masses with fractal correction (following section): $E_0 = \sqrt{m_e m_\mu / K_{\text{frac}}} = 7.398 \text{ MeV}$.

.6 Alternative Derivation of E_0 from Mass Ratios

The Geometric Mean of Lepton Energies

An alternative derivation of E_0 results directly from the geometric mean of electron and muon masses:

$$E_0 = \sqrt{m_e \cdot m_\mu} \cdot c^2 \quad (12)$$

With the masses calculated from quantum numbers:

$$E_0 = \sqrt{0.511 \text{ MeV} \times 105.66 \text{ MeV}} \quad (13)$$

$$= \sqrt{54.00 \text{ MeV}^2} \quad (14)$$

$$= 7.35 \text{ MeV} \quad (15)$$

Comparison with the Value Used

The value from the geometric mean (7.35 MeV) agrees well with the value used (7.398 MeV). The difference is less than 1%:

$$\Delta = \frac{7.398 - 7.35}{7.35} \times 100\% = 0.65\% \quad (16)$$

Physical Interpretation

That E_0 corresponds to the geometric mean of the lepton energies is interpreted physically as follows:

- E_0 represents a natural electromagnetic energy scale between electron and muon
- The relationship is purely geometric and requires no knowledge of α
- The mass ratio $m_\mu/m_e = 206.77$ is itself determined by quantum numbers

Precision Correction

The small difference between 7.35 MeV and 7.398 MeV is the fractal correction:

$$E_0^{\text{corrected}} = \frac{E_0^{\text{geom}}}{\sqrt{K_{\text{frac}}}} = \frac{7.348}{\sqrt{1 - 100\xi}} = \frac{7.348}{0.99331} = 7.397 \approx 7.398 \text{ MeV} \quad (17)$$

Verification of Fine Structure Constant

With the geometrically derived $E_0 = 7.35 \text{ MeV}$:

$$\varepsilon = \xi \cdot E_0^2 \quad (18)$$

$$= (1.333 \times 10^{-4}) \times (7.35)^2 \quad (19)$$

$$= (1.333 \times 10^{-4}) \times 54.02 \quad (20)$$

$$= 7.20 \times 10^{-3} \quad (21)$$

$$= \frac{1}{138.9} \quad (22)$$

With the fractally corrected value $E_0 = 7.348/\sqrt{K_{\text{frac}}}$ one gets $\alpha^{-1} = K_{\text{frac}}/(\xi \cdot 7.348^2) = 0.98667 \times 138.91 = 137.06$ (+0.017 %). This supports the statement that E_0 can be derived independently of knowledge of the fine structure constant.

The two routes to E_0 are the geometric mean without and with fractal correction, 7.348 and 7.397 MeV (previous section); conversely, the measured α fixes the factor $K_{\text{frac}} = 0.98650$, 1.7×10^{-4} from $1 - 100\xi$ (A130).

.7 The T0 Coupling Parameter ε

The T0 coupling parameter results as:

$$\varepsilon = \xi \cdot E_0^2 \quad (23)$$

With the derived values:

$$\varepsilon = (1.333 \times 10^{-4}) \times (7.398 \text{ MeV})^2 \quad (24)$$

$$= 7.297 \times 10^{-3} \quad (25)$$

$$= \frac{1}{137.036} \quad (26)$$

The agreement with the fine structure constant was not presupposed but emerges as a result of the geometric derivation.

The Simplest Formula for the Fine-Structure Constant

$$\alpha = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2$$

Important: The normalization (1 MeV)² is required for dimensionless results.

.8 Fractal Damping: No Independent Second Path

An apparent second path to α runs through a fractal damping:

$$\alpha_{\text{bare}}^{-1} = 3\pi \times \xi^{-1} \times \ln \left(\frac{\Lambda_{\text{Planck}}}{m_\mu} \right) \quad (27)$$

With the fractal damping factor:

$$D_{\text{frac}} = \left(\frac{\lambda_C^{(\mu)}}{\ell_P} \right)^{D_f - 2} \quad (28)$$

the result is supposed to be:

$$\alpha^{-1} = \alpha_{\text{bare}}^{-1} \times D_{\text{frac}} \quad (29)$$

This chain does not yield 137.036 by itself: $\lambda_C^{(\mu)}/\ell_P = 1.156 \times 10^{20}$; raised to $D_f - 2$ with $D_f = 3 - \xi$ this gives about 1.15×10^{20} . With $\alpha_{\text{bare}}^{-1} = 3\pi \times 7500 \times \ln(1.2209 \times 10^{22}/105.66) = 3\pi \times 7500 \times 46.196 = 3.265 \times 10^6$, D_{frac} would have to be exactly $137.036/3.265 \times 10^6 = 4.2 \times 10^{-5}$; this value is fitted to α (the exponent would have to be -0.218). This is not an independent derivation; the load-bearing relation is $\alpha = \xi E_0^2$ with $E_0 = 7.348/\sqrt{K_{\text{frac}}}$.

.9 Clarification: The Two Different κ Parameters

Important Distinction

In T0-theory literature, two physically different parameters are denoted by the symbol κ , which can lead to confusion. These must be clearly distinguished:

1. $\kappa_{\text{mass}} = 1.487$ - The fractal mass scaling exponent
2. κ_{grav} - The gravitational field parameter

The Mass Scaling Exponent κ_{mass}

This parameter was already derived in Section 4:

$$\kappa_{\text{mass}} = \frac{D_f^{\text{eff}}}{2} = 1.487 \quad (30)$$

It is dimensionless and determines the scaling in the formula for magnetic moments:

$$a_x \propto \left(\frac{m_x}{m_\mu} \right)^{\kappa_{\text{mass}}} \quad (31)$$

The Gravitational Field Parameter κ_{grav}

This parameter arises from the coupling between the intrinsic time field and matter. The T0 Lagrangian density reads:

$$\mathcal{L}_{\text{intrinsic}} = \frac{1}{2} \partial_\mu T \partial^\mu T - \frac{1}{2} T^2 - \frac{\rho}{T} \quad (32)$$

The resulting field equation:

$$\nabla^2 T = -\frac{\rho}{T^2} \quad (33)$$

leads to a modified gravitational potential:

$$\Phi(r) = -\frac{GM}{r} + \kappa_{\text{grav}} r \quad (34)$$

Relationship Between κ_{grav} and Fundamental Parameters

In natural units:

$$\kappa_{\text{grav}}^{\text{nat}} = \beta_T^{\text{nat}} \cdot \frac{y\nu}{r_g^2} \quad (35)$$

With $\beta_T = 1$ and $r_g = 2Gm_\mu$:

$$\kappa_{\text{grav}} = \frac{y_\mu \cdot \nu}{(2Gm_\mu)^2} = \frac{\sqrt{2}m_\mu \cdot \nu}{\nu \cdot 4G^2m_\mu^2} = \frac{\sqrt{2}}{4G^2m_\mu} \quad (36)$$

Numerical Value and Physical Significance

In SI units:

$$\kappa_{\text{grav}}^{\text{SI}} \approx 4.8 \times 10^{-11} \text{ m/s}^2 \quad (37)$$

This linear term in the gravitational potential:

- Is intended to explain observed flat rotation curves of galaxies [S]
- Would thereby make dark matter unnecessary [S]
- Arises naturally from time field-matter coupling

.10 Complete Mapping: Standard Model Parameters to T0 Correspondences

Overview of Parameter Reduction

The Standard Model requires over 20 free parameters that must be determined experimentally. The T0 system traces these back to the one parameter ξ , supplemented by integer structural choices (rational prefactors, exponents) that enter as motivated settings, and by reference anchors such as ν and E_p for unit conversion (for the classification of the early formulas, see the end of the document):

$$\xi = \frac{4}{3} \times 10^{-4} \quad (38)$$

Hierarchically Ordered Parameter Mapping Table

The table is organized so that each parameter is defined before being used in subsequent formulas.

On Table 1: the formulas $\Lambda_{QCD} = \nu \cdot \xi^{1/3}$ ($246 \times 0.05109 = 12.57 \text{ GeV}$) and $m_h = \nu \cdot \xi^{1/4}$ ($246 \times 0.10746 = 26.4 \text{ GeV}$) do not reproduce the measured values ($\sim 217 \text{ MeV}$ and 125.25 GeV); λ_h below them is computed with the measured $m_h = 125 \text{ GeV}$.

On Table 2: the coefficients $r_\tau = \frac{25}{9}$ and $r_c = 2$ correspond to Docs. 006 and 046. For m_s the formula with the factor 3 lies 5.4 % above 93.4 MeV; Doc. 006 uses $r_s = 26/9$ and hence 94.76 MeV. The neutrino values hold with $r \sim 1$: $\nu \xi^{3/2} \xi^3 = 9.0 \times 10^{-7} \text{ eV}$, $\nu \xi \xi^3 = 7.8 \times 10^{-5} \text{ eV}$, $\nu \xi^{2/3} \xi^3 = 1.5 \times 10^{-3} \text{ eV}$.

On Table 3: the formula for $|V_{us}|$ gives about 0.21 with the T0 masses (with the masses from Doc. 006: 0.213) and does not reproduce the SM value (-5%). For δ_{CKM} , $\arcsin(2\sqrt{2} \cdot \sqrt{\xi/3}) = 0.0109 \text{ rad}$; only without the factor $\sqrt{\xi}$ would one get $\arcsin(2\sqrt{2}/3) = 1.231 \text{ rad}$. For θ_{13} ,

Table 1: Standard Model Parameters in T0 Hierarchy — Part 1: Levels 0–3

SM Parameter	SM Value	T0 Formula	T0 Value
LEVEL 0: FUNDAMENTAL GEOMETRIC CONSTANT			
Geometric parameter ξ	–	$\xi = \frac{4}{3} \times 10^{-4}$ (from geometric)	1.333×10^{-4} (exact)
LEVEL 1: PRIMARY COUPLING CONSTANTS (dependent only on ξ)			
Strong coupling α_s	$\alpha_s \approx 0.118$ (at M_Z)	$\alpha_s = \xi^{-1/3}$ $= (1.333 \times 10^{-4})^{-1/3}$	19.57 (nat. units)
Weak coupling α_W	$\alpha_W \approx 1/30$	$\alpha_W = \xi^{1/2}$ $= (1.333 \times 10^{-4})^{1/2}$	1.15×10^{-2}
Gravitational coupling α_G	not in SM	$\alpha_G = \xi^2$ $= (1.333 \times 10^{-4})^2$	1.78×10^{-8}
Electromagnetic coupling	$\alpha = 1/137.036$	$\alpha_{EM} = 1$ (convention) $\varepsilon_T = \xi \cdot E_0^2$ (physical coupling)	1 7.297×10^{-3} (*see note)
LEVEL 2: ENERGY SCALES (dependent on ξ and Planck scale)			
Planck energy E_p	1.22×10^{19} GeV	Reference scale (from $G, \hbar, c$)	1.22×10^{19} GeV
Higgs-VEV v	246.22 GeV (theoretisch)	$v = \xi^4 E_p / (5\pi)$ (see appendix)	245.65 GeV
QCD scale Λ_{QCD}	~ 217 MeV (free parameter)	$\Lambda_{QCD} = v \cdot \xi^{1/3}$ $= 246 \text{ GeV} \cdot \xi^{1/3}$	12.6 GeV
LEVEL 3: HIGGS SECTOR (dependent on v)			
Higgs mass m_h	125.25 GeV (measured)	$m_h = v \cdot \xi^{1/4}$ $= 246 \cdot (1.333 \times 10^{-4})^{1/4}$	26.4 GeV
Higgs self-coupling λ_h	0.13 (derived)	$\lambda_h = \frac{m_h^2}{2v^2}$ $= \frac{(125)^2}{2(246)^2}$	0.129

Table 2: Standard Model Parameters in T0 Hierarchy — Part 2: Levels 4–5

SM Parameter	SM Value	T0 Formula	T0 Value
LEVEL 4: FERMION MASSES (dependent on v and ξ)			
<i>Leptons:</i>			
Electron mass m_e	0.511 MeV (free parameter)	$m_e = v \cdot \frac{4}{3} \cdot \xi^{3/2}$ $= 246 \text{ GeV} \cdot \frac{4}{3} \cdot \xi^{3/2}$	0.502 MeV
Muon mass m_μ	105.66 MeV (free parameter)	$m_\mu = v \cdot \frac{16}{5} \cdot \xi^1$ $= 246 \text{ GeV} \cdot \frac{16}{5} \cdot \xi$	105.0 MeV
Tau mass m_τ	1776.86 MeV (free parameter)	$m_\tau = v \cdot \frac{25}{9} \cdot \xi^{2/3}$ $= 246 \text{ GeV} \cdot \frac{25}{9} \cdot \xi^{2/3}$	1783 MeV
<i>Up-type quarks:</i>			
Up quark mass m_u	2.16 MeV	$m_u = v \cdot 6 \cdot \xi^{3/2}$	2.27 MeV
Charm quark mass m_c	1.27 GeV	$m_c = v \cdot 2 \cdot \xi^{2/3}$	1.284 GeV
Top quark mass m_t	172.76 GeV	$m_t = v \cdot \frac{1}{28} \cdot \xi^{-1/3}$	173.0 GeV
<i>Down-type quarks:</i>			
Down quark mass m_d	4.67 MeV	$m_d = v \cdot \frac{25}{2} \cdot \xi^{3/2}$	4.72 MeV
Strange quark mass m_s	93.4 MeV	$m_s = v \cdot 3 \cdot \xi^1$	98.4 MeV
Bottom quark mass m_b	4.18 GeV	$m_b = v \cdot \frac{3}{2} \cdot \xi^{1/2}$	4.254 GeV
LEVEL 5: NEUTRINO MASSES (dependent on v and double ξ)			
Electron neutrino m_{ν_e}	$< 2 \text{ eV}$ (upper limit)	$m_{\nu_e} = v \cdot r_{\nu_e} \cdot \xi^{3/2} \cdot \xi^3$ with $r_{\nu_e} \sim 1$	$\sim 10^{-6} \text{ eV}$ (prediction)
Muon neutrino m_{ν_μ}	$< 0.19 \text{ MeV}$	$m_{\nu_\mu} = v \cdot r_{\nu_\mu} \cdot \xi^1 \cdot \xi^3$	$\sim 10^{-4} \text{ eV}$
Tau neutrino m_{ν_τ}	$< 18.2 \text{ MeV}$	$m_{\nu_\tau} = v \cdot r_{\nu_\tau} \cdot \xi^{2/3} \cdot \xi^3$	$\sim 10^{-3} \text{ eV}$

Table 3: Standard Model Parameters in T0 Hierarchy — Part 3: Levels 6–7

SM Parameter	SM Value	T0 Formula	T0 Value
LEVEL 6: MIXING MATRICES (dependent on mass ratios)			
<i>CKM Matrix (Quarks):</i>			
$ V_{us} $ (Cabibbo)	0.22452	$ V_{us} = \sqrt{\frac{m_d}{m_s}} \cdot f_{Cab}$ with $f_{Cab} = \sqrt{\frac{m_s - m_d}{m_s + m_d}}$	0.21
$ V_{ub} $	0.00365	$ V_{ub} = \sqrt{\frac{m_d}{m_b}} \cdot \xi^{1/4}$	0.0037
$ V_{ud} $	0.97446	$ V_{ud} = \sqrt{1 - V_{us} ^2 - V_{ub} ^2}$ (unitarity)	0.974
CKM CP phase δ_{CKM}	1.20 rad	$\delta_{CKM} = \arcsin(2\sqrt{2}\xi^{1/2}/3)$	0.0109 rad
<i>PMNS Matrix (Neutrinos):</i>			
θ_{12} (Solar)	33.44°	$\theta_{12} = \arcsin \sqrt{m_{\nu_1}/m_{\nu_2}}$	33.5°
θ_{23} (Atmospheric)	49.2°	$\theta_{23} = \arcsin \sqrt{m_{\nu_2}/m_{\nu_3}}$	49°
θ_{13} (Reactor)	8.57°	$\theta_{13} = \arcsin(\xi^{1/3})$	2.93°
PMNS CP phase δ_{CP}	unknown	$\delta_{CP} = \pi(1 - 2\xi)$	3.14 rad
LEVEL 7: DERIVED PARAMETERS			
Weinberg $\sin^2 \theta_W$	0.2312	$\frac{1 - \sqrt{1 - 4\alpha_W}}{4}$ with α_W from Level 1	0.0058
Strong CP phase θ_{QCD}	$< 10^{-10}$ (upper limit)	$\theta_{QCD} = \xi^2$	1.78×10^{-8} (excluded)

$\arcsin(\xi^{1/3}) = 2.93^\circ$; 8.59° only follows with the argument $(25\xi)^{1/3}$, $25\xi = 1/300 = 1/(5|A_5|)$, the corpus formula of Doc. 328/340. The formula for $\sin^2 \theta_W$ gives 0.0058 with $\alpha_W = 0.01155$ and does not reproduce the SM value. The prediction $\theta_{QCD} = \xi^2 = 1.78 \times 10^{-8}$ lies about 178 times above the bound 10^{-10} from the electric dipole moment of the neutron quoted in the table and is therefore experimentally excluded.

The Hierarchical Derivation Structure

The table shows the clear hierarchy of parameter derivation:

1. **Level 0:** Only ξ as fundamental constant
 2. **Level 1:** Coupling constants directly from ξ
 3. **Level 2:** Energy scales from ξ and reference scales
 4. **Level 3:** Higgs parameters from energy scales
 5. **Level 4:** Fermion masses from ν and ξ
 6. **Level 5:** Neutrino masses with additional suppression
 7. **Level 6:** Mixing parameters from mass ratios
 8. **Level 7:** Further derived parameters
- Each level uses only parameters that were defined in previous levels.

Critical Notes

(*) Note on the Fine Structure Constant:

The fine structure constant has a dual function in the T0 system:

- $\alpha_{EM} = 1$ is a **unit convention** (like $c = 1$)
- $\varepsilon_T = \xi \cdot E_0^2 = 7.297 \times 10^{-3}$ is the **physical EM coupling**

Unit System: All T0 values apply in natural units with $\hbar = c = 1$. Transformation to SI units is required for experimental comparisons.

.11 Cosmological Parameters: Standard Cosmology (Λ CDM) vs T0 System

Basic Difference of the Approaches

Note: Basic Differences

The T0 system postulates a **static, eternal universe** without a Big Bang, while standard cosmology is based on an **expanding universe** with a Big Bang. The parameters are therefore often not directly comparable but represent different physical concepts. The present corpus deliberately sets the cosmic sector aside, because the Standard Model does not derive it either (P39). The cosmological T0 statements in this section are therefore speculative [S].

Table 4: Cosmological Parameters in Hierarchical Order — Part 1: Geometric & Energy Scales

Parameter	Λ CDM Value	T0 Formula	T0 Interpretation
LEVEL 0: FUNDAMENTAL GEOMETRIC CONSTANT			
Geometric parameter ξ	not present	$\xi = \frac{4}{3} \times 10^{-4}$ (from geometric)	1.333×10^{-4} basis of all deriva- tions
LEVEL 1: PRIMARY ENERGY SCALES (dependent only on ξ)			
Characteristic energy	–	$E_\xi = \frac{1}{\xi} = \frac{3}{4} \times 10^4$	7500 (nat. units) CMB energy scale
Characteristic length	–	$L_\xi = \xi$	1.33×10^{-4} (nat. units)
ξ -field energy density	–	$\rho_\xi = E_\xi^4$	3.16×10^{16} vacuum energy density
LEVEL 2: CMB PARAMETERS (dependent on ξ and E_ξ)			
CMB temperature today	$T_0 = 2.7255$ K (measured)	$T_{CMB} = \frac{16}{9} \xi^2 \cdot E_\xi$ $= \frac{16}{9} \cdot (1.33 \times 10^{-4})^2 \cdot 7500$	2.725 K (calculated)
CMB energy density	$\rho_{CMB} = 4.64 \times 10^{-31}$ kg/m ³	$\rho_{CMB} = \frac{\pi^2}{15} T_{CMB}^4$	4.2×10^{-14} J/m ³
CMB anisotropy	$\Delta T/T \sim 10^{-5}$ (Planck satellite)	Stefan-Boltzmann $\delta T = \xi^{1/2} \cdot T_{CMB}$ quantum fluctuation	(nat. units) $\sim 10^{-5}$ (predicted)

Table 5: Cosmological Parameters — Part 2: Redshift & Dark Components

Parameter	Λ CDM Value	T0 Formula	T0 Interpretation
LEVEL 3: REDSHIFT (dependent on ξ and wavelength)			
Redshift z	(Planck 2020)	Static universe	Fractal path- lengthening not expansion
	$z = \frac{\Delta\lambda}{\lambda}$ (expansion)	$z(\lambda, d) = \xi \cdot \lambda \cdot d$ Wavelength- dependent (original state; cf. K6)	
Effective H_0 (interpreted)	66.2 km/s/Mpc	see Document 026 at $\lambda_{ref} = 550$ nm	(apparent)
LEVEL 4: DARK COMPONENTS			
Dark energy Ω_Λ	0.6847 ± 0.0073 (68.47% of uni- verse)	Not required Static universe	0 eliminated
Dark matter Ω_{DM}	0.2607 ± 0.0067 (26.07% of uni- verse)	ξ -field effects Modified gravity	0 eliminated
Baryonic matter Ω_b	0.0492 ± 0.0003 (4.92% of uni- verse)	All matter	1.0 (100%)
Cosmological constant Λ	$(1.1 \pm 0.02) \times 10^{-52}$ m ⁻²	$\Lambda = 0$	0
		No expansion	eliminated

Table 6: Cosmological Parameters — Part 3: Universe Structure, Predictions & Energy Balances

Parameter	Λ CDM Value	T0 Formula	T0 Interpretation
LEVEL 5: UNIVERSE STRUCTURE			
Universe age	13.787 ± 0.020 Gyr (since Big Bang)	$t_{univ} = \infty$ No beginning/end	Eternal Static
Big Bang	$t = 0$ Singularity	No Big Bang Excluded by $T \cdot E = 1$ (Doc. 306)	– Excluded
Decoupling (CMB)	$z \approx 1100$ (Λ CDM) $t = 380,000$ years	CMB from ξ -field Vacuum fluctuation	Continuous generation
Structure formation	Bottom-up (small $\rightarrow$ large)	Continuous ξ -driven	Cyclic regenerating
LEVEL 6: DISTINGUISHABLE PREDICTIONS			
Hubble tension	Unsolved $H_0^{local} \neq H_0^{CMB}$	Interpreted via ξ -effects	No tension [S] $H_0^{eff} = 67.45$
JWST early galaxies	Problem (formed too early)	No problem Eternal universe	Expected in static universe
λ -dependent z	z independent of λ All λ same z	$z \propto \lambda$ $z_{UV} > z_{radio}$	At the limit of testability*
Casimir effect	Quantum fluctu- ation	$F_{Cas} = -\frac{\pi^2}{240} \frac{\hbar c}{d^4}$ from ξ -geometry	ξ -field manifestation
LEVEL 7: ENERGY BALANCES			
Total energy	Not conserved (expansion)	$E_{total} = const$	Strictly conserved
Mass-energy equivalence	$E = mc^2$	$E = mc^2$	Identical** (see note)
Vacuum energy	Problem (10^{120} discrep- ancy)	$\rho_{vac} = \rho_\xi$ Calculable from ξ	Naturally from ξ
Entropy	Grows monoton- ically (heat death)	$S_{total} = const$ Regeneration	Cyclically conserved

Phenomenon	Λ CDM Explanation	T0 Interpretation [S]
Redshift	Space expansion	Fractal path-lengthening in the ξ -field (geometric path stretching; no photon energy loss)
CMB	Recombination at $z = 1100$ (Λ CDM value)	ξ -field equilibrium radiation
Dark energy	68% of universe	Not required
Dark matter	26% of universe	ξ -field gravity effects
Hubble tension	Unsolved (4.4σ)	Interpreted via ξ -effects
JWST paradox	Unexplained early galaxies	No problem in eternal universe

Table 7: Differences between Λ CDM and T0 (T0 column speculative, cf. P39)

Hierarchically Ordered Cosmological Parameters

Critical Differences and Test Possibilities

Philosophical Implications

In the T0 picture, the following would follow [S]:

1. **Eternal universe:** No beginning, no end – the question of a first beginning does not arise
2. **No singularities:** A minimal length scale L_0 (set by ξ) replaces the formal Big Bang singularity by a tiny but finite core; the justification via Heisenberg uncertainty has been replaced by $T \cdot E = 1$ (Doc. 306)
3. **Energy conservation:** Strictly preserved, no violation through expansion
4. **Simplicity:** One constant instead of 6+ parameters (the cosmological sector still requires its own scale, P39)
5. **Testability:** Clear, measurable predictions

.12 Appendix: Purely Theoretical Derivation of Higgs VEV from Quantum Numbers

Fundamental theoretical foundations

Quantum numbers of leptons in T0 theory

T0 theory assigns quantum numbers (n, l, j) to each particle, arising from the solution of the three-dimensional wave equation in the energy field:

Electron (1st generation):

- Principal quantum number: $n = 1$
- Orbital angular momentum: $l = 0$ (s-like, spherically symmetric)
- Total angular momentum: $j = 1/2$ (fermion)

Muon (2nd generation):

- Principal quantum number: $n = 2$
- Orbital angular momentum: $l = 1$ (p-like, dipole structure)

- Total angular momentum: $j = 1/2$ (fermion)

Universal mass formulas

T0 theory uses two formulations for particle masses; as the mass ratios below show, they are not equivalent:

Direct method:

$$m_i = \frac{1}{\xi_i} = \frac{1}{\xi_0 \times f(n_i, l_i, j_i)} \quad (39)$$

Extended Yukawa method:

$$m_i = y_i \times v \quad (40)$$

where:

- $\xi_0 = \frac{4}{3} \times 10^{-4}$: Universal geometric parameter
- $f(n_i, l_i, j_i)$: Geometric factors from quantum numbers
- y_i : Yukawa couplings
- v : Higgs VEV (target quantity)

Theoretical calculation of geometric factors

Geometric factors from quantum numbers

The geometric factors result from the analytical solution of the three-dimensional wave equation. For the fundamental leptons:

Electron ($n = 1, l = 0, j = 1/2$):

The ground state solution of the 3D wave equation yields the simplest geometric factor:

$$f_e(1, 0, 1/2) = 1 \quad (41)$$

This is the reference configuration (ground state).

Muon ($n = 2, l = 1, j = 1/2$):

For the first excited configuration with dipole character, the solution yields:

$$f_\mu(2, 1, 1/2) = \frac{16}{5} \quad (42)$$

This factor accounts for:

- $n^2 = 4$ (energy level scaling)
- $\frac{4}{5}$ ($l = 1$ dipole correction vs. $l = 0$ spherical)

Verification of factors

The geometric factors must be consistent with the universal T0 structure:

$$\xi_e = \xi_0 \times f_e = \frac{4}{3} \times 10^{-4} \times 1 = \frac{4}{3} \times 10^{-4} \quad (43)$$

$$\xi_\mu = \xi_0 \times f_\mu = \frac{4}{3} \times 10^{-4} \times \frac{16}{5} = \frac{64}{15} \times 10^{-4} \quad (44)$$

Derivation of mass ratios

Theoretical electron-muon mass ratio

With the geometric factors, it follows from the direct method:

$$\frac{m_\mu}{m_e} = \frac{\xi_e}{\xi_\mu} = \frac{f_e}{f_\mu} = \frac{1}{\frac{16}{5}} = \frac{5}{16} \quad (45)$$

Note: This is the inverse ratio. Since $\xi \propto 1/m$, we obtain:

$$\frac{m_\mu}{m_e} = \frac{f_\mu}{f_e} = \frac{\frac{16}{5}}{1} = \frac{16}{5} = 3.2 \quad (46)$$

The direct method thus gives $m_\mu/m_e = 16/5 = 3.2$, the Yukawa method below gives 207.8; the two formulations yield different ratios and are not equivalent. In addition, the muon is assigned $l = 1$ here but $l = 0$ in the section "Lepton Masses from Quantum Numbers".

Correction through Yukawa couplings

The Yukawa method accounts for additional quantum field theoretical corrections:

Electron:

$$y_e = \frac{4}{3} \times \xi^{3/2} = \frac{4}{3} \times \left(\frac{4}{3} \times 10^{-4}\right)^{3/2} \quad (47)$$

Muon:

$$y_\mu = \frac{16}{5} \times \xi^1 = \frac{16}{5} \times \frac{4}{3} \times 10^{-4} \quad (48)$$

Calculation of corrected ratio

$$\frac{y_\mu}{y_e} = \frac{\frac{16}{5} \times \frac{4}{3} \times 10^{-4}}{\frac{4}{3} \times \left(\frac{4}{3} \times 10^{-4}\right)^{3/2}} \quad (49)$$

$$= \frac{\frac{16}{5} \times \frac{4}{3} \times 10^{-4}}{\frac{4}{3} \times \frac{4}{3} \times 10^{-4} \times \sqrt{\frac{4}{3} \times 10^{-4}}} \quad (50)$$

$$= \frac{\frac{16}{5}}{\frac{4}{3} \times \sqrt{\frac{4}{3} \times 10^{-4}}} \quad (51)$$

$$= \frac{\frac{16}{5}}{\frac{4}{3} \times 0.01155} \quad (52)$$

$$= \frac{3.2}{0.0154} = 207.8 \quad (53)$$

This theoretical ratio of 207.8 is very close to the experimental value of 206.768.

Higgs VEV

In the corpus the Higgs VEV follows from the structural formula

$$\frac{v}{E_P} = \frac{\xi^4}{5\pi}, \quad v = 245.65 \text{ GeV} \quad (-0.23 \% \text{ against } 246.22 \text{ GeV}), \quad (54)$$

dimensionless; the factor 10 in $5\pi = (\pi/2) \cdot 10$ is fitted to the measured value, and with the bare v of the ladder it becomes $10 K_{\text{frak}} \approx \pi^2$ (Docs. 149, 387). The mass ratio $m_\mu/m_e = 207.8$ (0.5 %) from the previous section does not contain v and is unaffected.

.13 List of Symbols Used

Fundamental Constants

Symbol	Meaning	Value/Unit
ξ	Geometric parameter	$\frac{4}{3} \times 10^{-4}$ (dimensionless)
c	Speed of light	2.998×10^8 m/s
$\hbar$	Reduced Planck constant	1.055×10^{-34} J·s
G	Gravitational constant	6.674×10^{-11} m ³ /(kg·s ²)
k_B	Boltzmann constant	1.381×10^{-23} J/K
e	Elementary charge	1.602×10^{-19} C

Coupling Constants

Symbol	Meaning	Formula
α	Fine structure constant	$1/137.036$ (SI)
α_{EM}	Electromagnetic coupling	1 (nat. units)
α_S	Strong coupling	$\xi^{-1/3}$
α_W	Weak coupling	$\xi^{1/2}$
α_G	Gravitational coupling	ξ^2
ε_T	T0 coupling parameter	$\xi \cdot E_0^2$

Energy Scales and Masses

Symbol	Meaning	Value/Formula
E_P	Planck energy	1.22×10^{19} GeV
E_ξ	Characteristic energy	$1/\xi = 7500$ (nat. units)
E_0	Fundamental EM energy	7.398 MeV
v	Higgs VEV	246.22 GeV
m_h	Higgs mass	125.25 GeV
Λ_{QCD}	QCD scale	~ 200 MeV
m_e	Electron mass	0.511 MeV
m_μ	Muon mass	105.66 MeV
m_τ	Tau mass	1776.86 MeV
m_u, m_d	Up, down quark masses	2.16, 4.67 MeV
m_c, m_s	Charm, strange quark masses	1.27 GeV, 93.4 MeV
m_t, m_b	Top, bottom quark masses	172.76 GeV, 4.18 GeV
$m_{\nu_e}, m_{\nu_\mu}, m_{\nu_\tau}$	Neutrino masses	< 2 eV, < 0.19 MeV, < 18.2 MeV

Cosmological Parameters

Symbol	Meaning	Value/Formula
T_{CMB}	CMB temperature	2.725 K
z	Redshift	dimensionless
Ω_Λ	Dark energy density	0.6847 (Λ CDM), 0 (T0)
Ω_{DM}	Dark matter density	0.2607 (Λ CDM), 0 (T0)
Ω_b	Baryon density	0.0492 (Λ CDM), 1 (T0)
Λ	Cosmological constant	$(1.1 \pm 0.02) \times 10^{-52} \text{ m}^{-2}$
ρ_ξ	ξ -field energy density	E_ξ^4
ρ_{CMB}	CMB energy density	$4.64 \times 10^{-31} \text{ kg/m}^3$

Geometric and Derived Quantities

Symbol	Meaning	Value/Formula
D_f^{eff}	Effective fractal dimension (cumulative over RG running)	2.973
κ_{mass}	Mass scaling exponent	$D_f^{\text{eff}}/2 = 1.487$
κ_{grav}	Gravitational field parameter	$4.8 \times 10^{-11} \text{ m/s}^2$
λ_h	Higgs self-coupling	0.13
θ_W	Weinberg angle	$\sin^2 \theta_W = 0.2312$
θ_{QCD}	Strong CP phase	$< 10^{-10} \text{ (exp.)}$, ξ^2 (T0, excluded)
ℓ_P	Planck length	$1.616 \times 10^{-35} \text{ m}$
λ_C	Compton wavelength	$\hbar/(mc)$
r_g	Gravitational radius	$2Gm$
L_ξ	Characteristic length	ξ (nat. units)

Mixing Matrices

Symbol	Meaning	Typical Value
V_{ij}	CKM matrix elements	see table
$ V_{ud} $	CKM ud element	0.97446
$ V_{us} $	CKM us element (Cabibbo)	0.22452
$ V_{ub} $	CKM ub element	0.00365
δ_{CKM}	CKM CP phase	1.20 rad
θ_{12}	PMNS solar angle	33.44°
θ_{23}	PMNS atmospheric	49.2°
θ_{13}	PMNS reactor angle	8.57°
δ_{CP}	PMNS CP phase	unknown

Other Symbols

Symbol	Meaning	Context
n, l, j	Quantum numbers	Particle classification
r_i	Rational coefficients	Yukawa couplings
p_i	Generation exponents	$3/2, 1, 2/3, \dots$
$f(n, l, j)$	Geometric function	Mass formula
ρ_{tet}	Tetrahedral packing density	0.68
γ	Universal exponent	1.01
ν	Crystal symmetry factor	0.63
β_T	Time field coupling	1 (nat. units)
y_i	Yukawa couplings	$r_i \cdot \xi^{p_i}$
$T(x, t)$	Time field	T0 theory
E_{field}	Energy field	Universal field

Classification of the Early Coupling-Constant Formulas

The early coupling-constant formulas developed in this document are to be classified as preliminary work, not as canonical FFGFT derivations. Authoritative documents are Doc. 160 (lepton lifetime structure), Doc. 318 (validity range of mass derivations), and the Galois series (Docs. 336–348). Numerical use of the formulas in this document requires recourse to these successor documents.